

Customer Engagement & Product Utilization Analytics for Retention Strategy

Internship Project Report

Submitted By: Rajshree Verma

Course: MBA (Finance & HR)

Internship Role: Finance Analyst Intern

Organization: Unified Mentor

Project Domain: Customer Retention Analytics and Banking Analytics

Submission Date: June 2026

Project Objective

To analyze customer engagement, product utilization, and relationship strength to identify factors influencing customer retention and churn. The project aims to provide data-driven insights for designing effective retention strategies, improving customer loyalty, and enhancing product adoption within the banking sector.

✓ KPI Summary

KPI	Result
Total Customers	10,000
Churn Rate	20.37%
Retention Rate	79.63%
Active Customer Churn Rate	14.27%
Inactive Customer Churn Rate	26.85%
High-Balance Disengagement Rate	49.88%
Germany Churn Rate	32.44%
France Churn Rate	16.15%
Spain Churn Rate	16.67%
Average RSI (Retained Customers)	0.518
Average RSI (Churned Customers)	0.432
Best Product Count for Retention	2 Products
Retention Rate (2 Products)	92.42%
Average Age of Retained Customers	37.41 Years
Average Age of Churned Customers	44.84 Years

KPI	Result
Average Balance of Retained Customers	72,745
Average Balance of Churned Customers	91,109
Credit Card Impact on Retention	Minimal

Introduction

Customer retention is a critical challenge in the banking industry. While traditional retention strategies often focus on demographic and financial indicators, customer engagement and product utilization provide deeper insights into customer loyalty and churn behavior.

This study analyzes customer engagement patterns, product adoption levels, account balances, and customer activity to identify factors influencing retention. The analysis aims to support data-driven retention strategies by understanding how customer behavior impacts long-term relationships with the bank.

Problem Statement

Banks often possess extensive customer data but lack clear insights into the behavioral factors that drive customer retention.

High account balances and strong financial profiles do not always guarantee loyalty. Customers may churn due to low engagement, limited product utilization, or weak relationships with the bank.

This project aims to identify the key behavioral indicators associated with churn and provide actionable recommendations to improve retention.

Objectives

Primary Objectives

- Evaluate the relationship between customer engagement and churn.
- Measure the retention impact of product count and product utilization.
- Identify disengaged yet high-value customers.

Secondary Objectives

- Support engagement-driven retention strategies.
- Improve product bundling and cross-selling decisions.
- Reduce churn among premium customers.

Executive Summary

This project analyzes customer engagement and product utilization to understand the key factors influencing customer retention in the banking sector. Using customer demographic and banking data, the analysis identifies patterns related to customer activity, product usage, account balance, and geographical distribution.

An interactive dashboard was developed using Python, Pandas, and Streamlit to present key performance indicators and visualize customer churn trends. The analysis shows that active customers and customers using an appropriate number of banking products generally have better retention. It also highlights differences in churn across geographical regions, indicating the need for targeted customer retention strategies.

The findings can support management in making informed decisions regarding customer engagement, product offerings, and retention initiatives. By focusing on customer behavior and service utilization, banks can improve customer satisfaction, reduce churn, and strengthen long-term relationships. This project demonstrates how data-driven analysis can assist organizations in developing effective customer retention strategies.

✓ Dataset Overview

For this project, a customer dataset containing information of 10,000 bank customers was analyzed. The dataset includes customer demographics, banking relationship details, financial information, product usage, and customer activity status.

The purpose of using this dataset is to understand customer behavior and identify the factors that influence customer retention and customer churn.

The dataset contains customers from three countries: France, Germany, and Spain. Information such as age, credit score, account balance, number of products used, credit card ownership, activity status, and estimated salary has been considered during the analysis.

The target variable in the dataset is "Exited", which indicates whether a customer has left the bank or remained with the bank.

Dataset Highlights

- Total Customers Analyzed: 10,000
- Countries Covered: France, Germany, and Spain
- Average Customer Age: 38.92 years
- Average Account Balance: 76,485.89
- Average Credit Score: 650.53
- Average Products Used: 1.53
- Active Members: 51.51%

- Customers with Credit Cards: 70.55%
- Overall Churn Rate: 20.37%

The dataset provides useful insights into customer engagement and product utilization, which can help banks develop better retention strategies and improve customer relationships.

✓ Dataset Structure

The dataset was examined to understand the available variables, data types, and completeness of information. The output below shows the dataset structure and confirms that there are no missing values.

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 14 columns):
 #   Column                Non-Null Count  Dtype  
---  -
 0   Year                  10000 non-null  int64  
 1   CustomerId            10000 non-null  int64  
 2   Surname                10000 non-null  object  
 3   CreditScore            10000 non-null  int64  
 4   Geography              10000 non-null  object  
 5   Gender                 10000 non-null  object  
 6   Age                   10000 non-null  int64  
 7   Tenure                 10000 non-null  int64  
 8   Balance                10000 non-null  float64 
 9   NumOfProducts          10000 non-null  int64  
10   HasCrCard              10000 non-null  int64  
11   IsActiveMember         10000 non-null  int64  
12   EstimatedSalary        10000 non-null  float64 
13   Exited                 10000 non-null  int64  
dtypes: float64(2), int64(9), object(3)
memory usage: 1.1+ MB
```

✓ Summary Statistics

Descriptive statistics were generated to understand the distribution and average values of key variables such as age, balance, credit score, tenure, and estimated salary.

```
df.describe()
```

	Year	CustomerId	CreditScore	Age	Tenure	Balance
count	10000.0	1.000000e+04	10000.000000	10000.000000	10000.000000	10000.000000
mean	2025.0	1.569094e+07	650.528800	38.921800	5.012800	76485.800000
std	0.0	7.193619e+04	96.653299	10.487806	2.892174	62397.400000
min	2025.0	1.556570e+07	350.000000	18.000000	0.000000	0.000000
25%	2025.0	1.562853e+07	584.000000	32.000000	3.000000	0.000000
50%	2025.0	1.569074e+07	652.000000	37.000000	5.000000	97198.500000
75%	2025.0	1.575323e+07	718.000000	44.000000	7.000000	127644.200000
max	2025.0	1.581569e+07	850.000000	92.000000	10.000000	250898.000000

Methodology

The project follows a structured analytical approach:

1. Data Loading and Validation

- Imported the customer dataset.
- Verified data quality and completeness.

2. Exploratory Data Analysis

- Examined customer demographics.
- Analyzed churn distribution.

3. Engagement Analysis

- Compared churn rates between active and inactive customers.

4. Product Utilization Analysis

- Evaluated churn across different product counts.

5. Customer Segmentation

- Identified Sticky Customers, Cross-Sell Opportunities, High Churn Risk Customers, and At-Risk Premium Customers.

6. KPI Development

- Engagement Retention Ratio
- Relationship Strength Index
- High-Balance Disengagement Rate
- Credit Card Stickiness Score

Exploratory Data Analysis (EDA)

```
card_churn = pd.crosstab(
    df['HasCrCard'],
    df['Exited'],
```

```

        normalize='index'
    )*100

card_churn

```

```

-----
NameError                                Traceback (most recent call last)
/tmp/ipykernel_1565/1640319311.py in <cell line: 0>()
      1 card_churn = pd.crosstab(
----> 2     df['HasCrCard'],
      3     df['Exited'],
      4     normalize='index'
      5 )*100

NameError: name 'df' is not defined

```

```

card_churn = pd.crosstab(
    df['HasCrCard'],
    df['Exited'],
    normalize='index'
) * 100

card_churn

```

	Exited	0	1
HasCrCard			
0	79.185059	20.814941	
1	79.815734	20.184266	

```

age_churn = df.groupby('Exited')['Age'].mean()

age_churn

```

	Age
Exited	
0	37.408389
1	44.837997

dtype: float64

```

balance_churn = df.groupby('Exited')['Balance'].mean()

balance_churn

```

Balance	
Exited	
0	72745.296779
1	91108.539337
dtype: float64	

```
print("Churned Customers:", df['Exited'].sum())
```

Churned Customers: 2037

```
geo_churn = pd.crosstab(
    df['Geography'],
    df['Exited'],
    normalize='index'
)*100
```

```
print(geo_churn)
```

Exited	0	1
Geography		
France	83.845233	16.154767
Germany	67.556796	32.443204
Spain	83.326605	16.673395

✓ KPI Analysis

To evaluate customer retention and relationship strength, several Key Performance Indicators (KPIs) were developed and analyzed. These KPIs helped in understanding how customer engagement, product utilization, and financial commitment influence customer loyalty.

✓ 1. Engagement Retention Ratio

This KPI compares the churn rates of active and inactive customers. The objective is to understand whether customer engagement influences customer retention.

```
active_retention = 85.730926
inactive_retention = 73.149103

ERR = active_retention / inactive_retention

print("Engagement Retention Ratio:", round(ERR,2))
```

Engagement Retention Ratio: 1.17

```
retention_rate = (1-df['Exited'].mean())*100

print("Retention Rate:", round(retention_rate,2), "%")
```

Retention Rate: 79.63 %

2. Product Depth Index

The Product Depth Index was assessed using the number of banking products utilized by customers. Customers using two products showed the highest retention rate, suggesting that greater product adoption strengthens the customer-bank relationship.

```
product_churn = pd.crosstab(
    df['NumOfProducts'],
    df['Exited'],
    normalize='index'
)*100

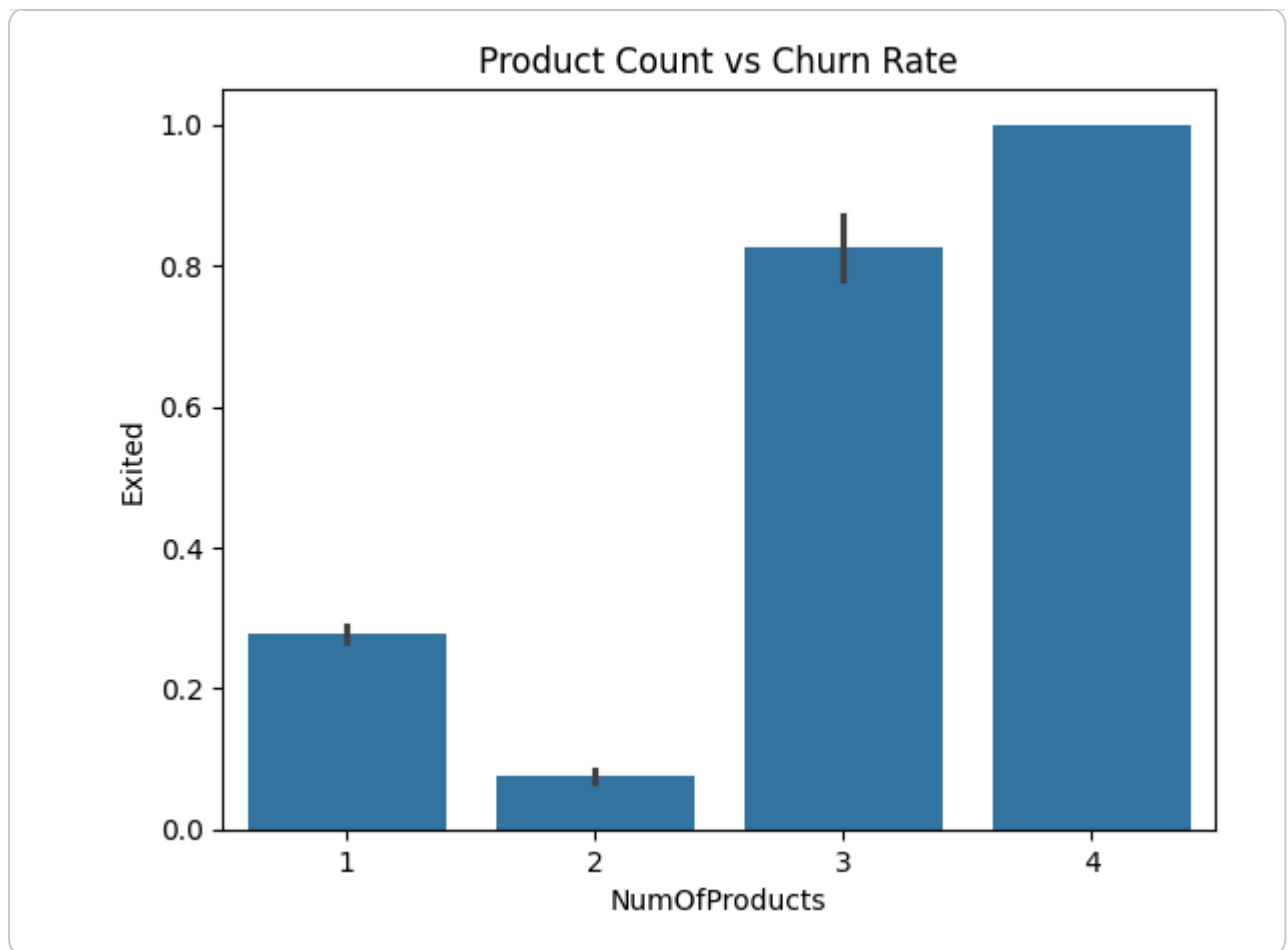
print(product_churn)
```

Exited	0	1
NumOfProducts		
1	72.285602	27.714398
2	92.418301	7.581699
3	17.293233	82.706767
4	0.000000	100.000000

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.barplot(
    x='NumOfProducts',
    y='Exited',
    data=df
)

plt.title("Product Count vs Churn Rate")
plt.show()
```

3. High-Balance Disengagement Rate

This KPI was used to identify customers who maintain high account balances but remain inactive. The findings showed that a substantial proportion of premium customers were disengaged, highlighting a potential risk of silent churn.

```
high_balance = df['Balance'].quantile(0.75)

premium_customers = df[df['Balance'] >= high_balance]

at_risk_premium = premium_customers[
    premium_customers['IsActiveMember'] == 0
]

print("Premium Customers:", len(premium_customers))
print("At-Risk Premium Customers:", len(at_risk_premium))
```

```
Premium Customers: 2500
At-Risk Premium Customers: 1247
```

```
risk_rate = len(at_risk_premium) / len(premium_customers) * 100

print("High-Balance Disengagement Rate:", round(risk_rate,2), "%")
```

```
High-Balance Disengagement Rate: 49.88 %
```

4. Credit Card Stickiness Score

The impact of credit card ownership on retention was evaluated. The results indicated only a marginal difference in churn rates between customers with and without credit cards, suggesting that card ownership alone is not a strong retention driver.

```
card_churn = pd.crosstab(
    df['HasCrCard'],
    df['Exited'],
    normalize='index'
)*100
```

```
card_churn
```

	Exited	0	1
HasCrCard			
0	79.185059	20.814941	
1	79.815734	20.184266	

5. Relationship Strength Index (RSI)

A Relationship Strength Index was created by combining customer activity status and product utilization levels. Customers with higher RSI scores demonstrated lower churn rates, indicating stronger relationships with the bank and higher retention potential.

```
df['RSI'] = (
    0.4 * df['IsActiveMember']
    + 0.4 * (df['NumOfProducts'] / 4)
    + 0.2 * df['HasCrCard']
)
```

```
df['RSI'].describe()
```

	RSI
count	10000.000000
mean	0.500160
std	0.226908
min	0.100000
25%	0.300000
50%	0.500000
75%	0.700000
max	1.000000

dtype: float64

Overall, the KPI analysis confirms that customer engagement and product utilization are stronger indicators of retention than financial value alone. These findings can support the development of targeted retention strategies and customer relationship management initiatives.

Customer Segmentation

```
df['CustomerSegment'] = 'Other'

# Sticky Customers
df.loc[
    (df['IsActiveMember']==1) &
    (df['NumOfProducts']>=2),
    'CustomerSegment'
] = 'Sticky Customer'

# At-Risk Premium
df.loc[
    (df['IsActiveMember']==0) &
    (df['Balance']>=df['Balance'].quantile(0.75)),
    'CustomerSegment'
] = 'At-Risk Premium'

# Cross-Sell Opportunity
df.loc[
    (df['IsActiveMember']==1) &
    (df['NumOfProducts']==1),
    'CustomerSegment'
] = 'Cross-Sell Opportunity'

# High Churn Risk
df.loc[
    (df['IsActiveMember']==0) &
```

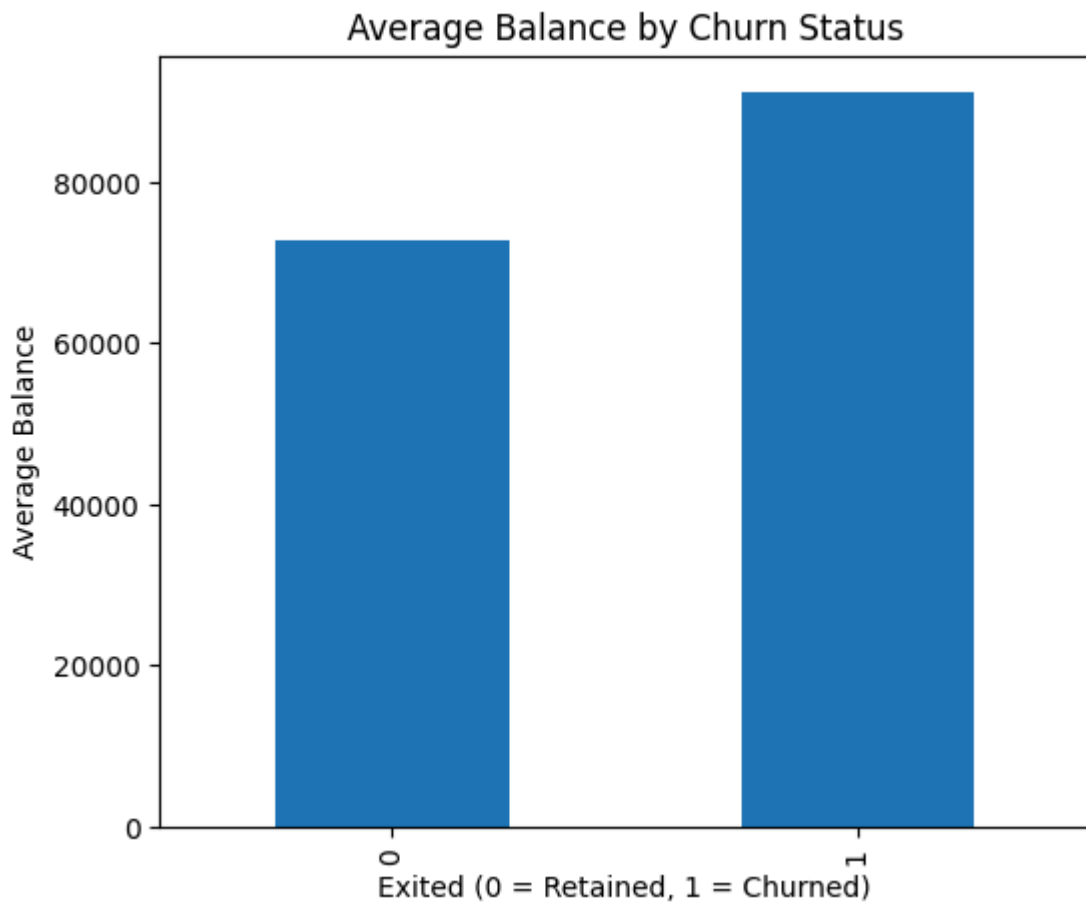
```
(df['NumOfProducts']==1),  
'CustomerSegment'  
] = 'High Churn Risk'  
  
df['CustomerSegment'].value_counts()
```

CustomerSegment	count
Sticky Customer	2588
Cross-Sell Opportunity	2563
High Churn Risk	2521
Other	1909
At-Risk Premium	419

dtype: int64

Visualizations

```
balance_churn = df.groupby('Exited')['Balance'].mean()  
  
import matplotlib.pyplot as plt  
  
plt.figure(figsize=(6,5))  
  
balance_churn.plot(kind='bar')  
  
plt.title("Average Balance by Churn Status")  
plt.xlabel("Exited (0 = Retained, 1 = Churned)")  
plt.ylabel("Average Balance")  
  
plt.show()
```



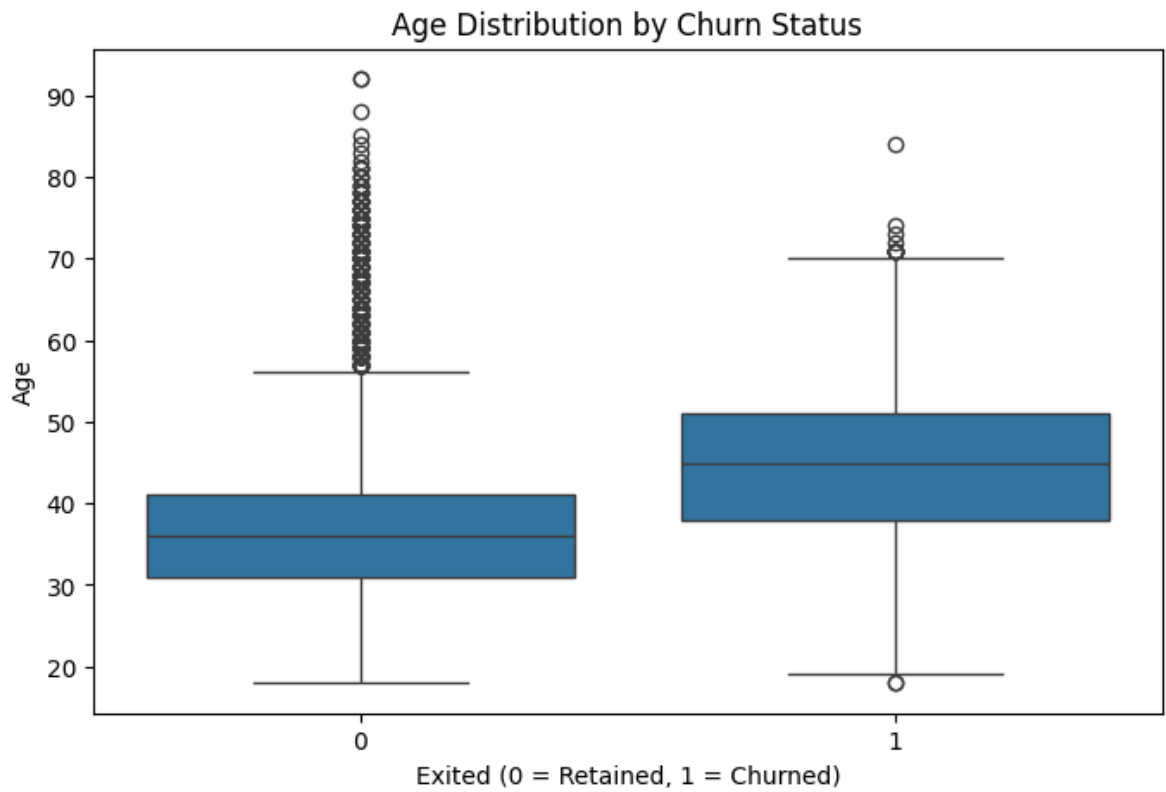
```
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(8,5))

sns.boxplot(x='Exited', y='Age', data=df)

plt.title("Age Distribution by Churn Status")
plt.xlabel("Exited (0 = Retained, 1 = Churned)")
plt.ylabel("Age")

plt.show()
```



```
geo_churn = df.groupby('Geography')['Exited'].mean()*100

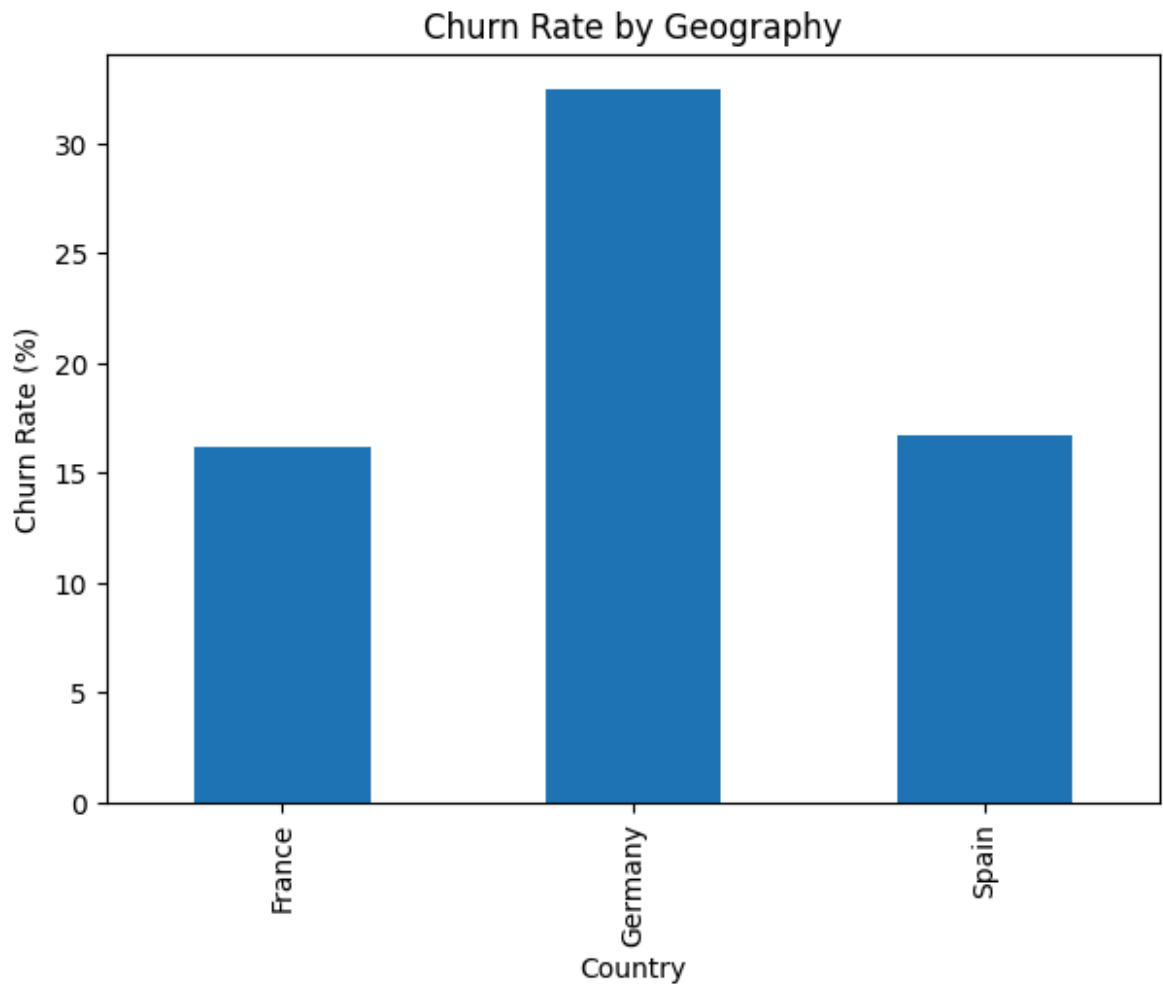
import matplotlib.pyplot as plt

plt.figure(figsize=(7,5))

geo_churn.plot(kind='bar')

plt.title("Churn Rate by Geography")
plt.xlabel("Country")
plt.ylabel("Churn Rate (%)")

plt.show()
```



```
product_churn_rate = df.groupby('NumOfProducts')['Exited'].mean()*100

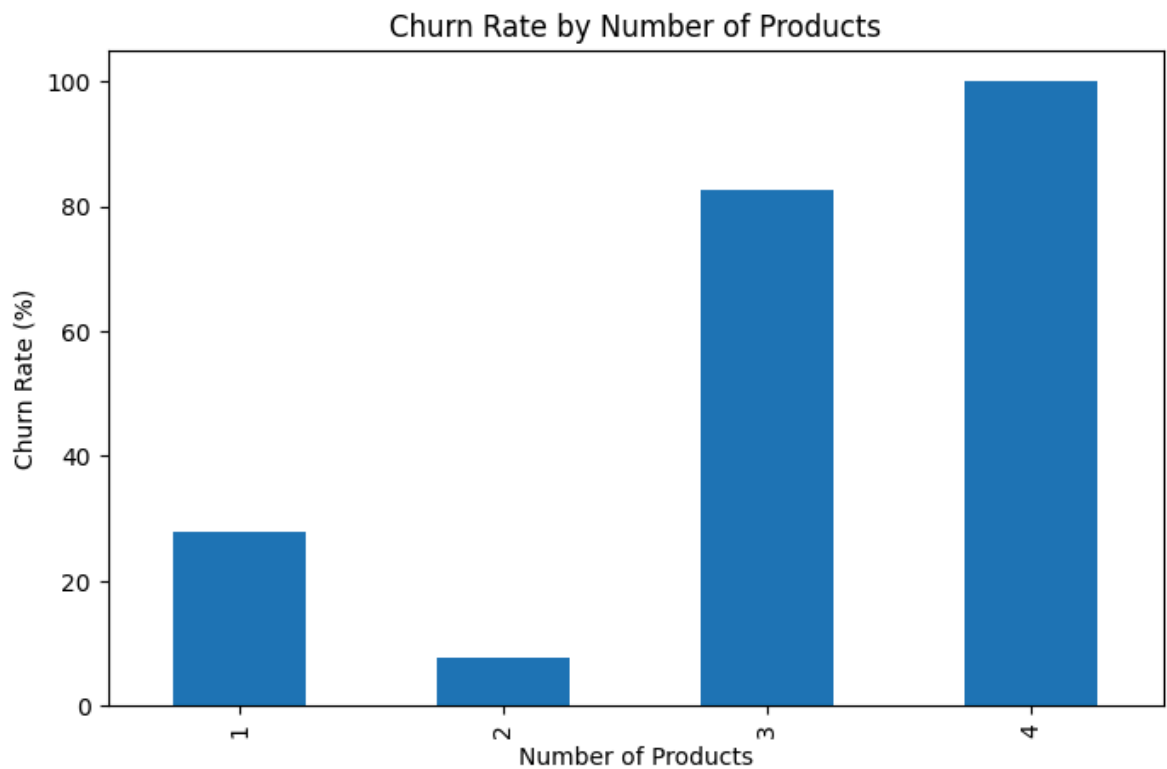
import matplotlib.pyplot as plt

plt.figure(figsize=(8,5))

product_churn_rate.plot(kind='bar')

plt.title("Churn Rate by Number of Products")
plt.xlabel("Number of Products")
plt.ylabel("Churn Rate (%)")

plt.show()
```



```
churn_rate = df['Exited'].mean()*100
```

```
print("Churn Rate:", round(churn_rate,2), "%")
```

```
Churn Rate: 20.37 %
```

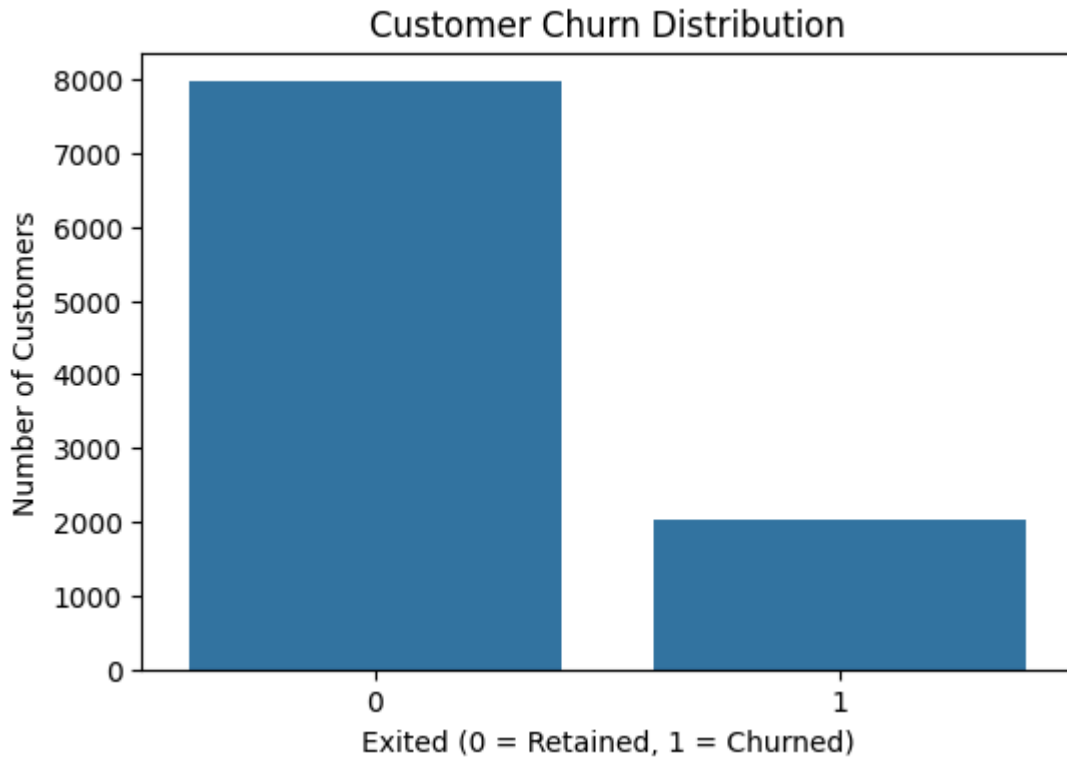
```
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(6,4))

sns.countplot(x='Exited', data=df)

plt.title("Customer Churn Distribution")
plt.xlabel("Exited (0 = Retained, 1 = Churned)")
plt.ylabel("Number of Customers")

plt.show()
```

```
import pandas as pd

df = pd.read_csv("European_Bank.csv")

df.head()
```

	Year	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	I
0	2025	15634602	Hargrave	619	France	Female	42	2	
1	2025	15647311	Hill	608	Spain	Female	41	1	8
2	2025	15619304	Onio	502	France	Female	42	8	15
3	2025	15701354	Boni	699	France	Female	39	1	
4	2025	15737888	Mitchell	850	Spain	Female	43	2	12

```
df.shape
```

```
(10000, 14)
```

```
df.isnull().sum()
```

	0
Year	0
CustomerId	0
Surname	0
CreditScore	0
Geography	0
Gender	0
Age	0
Tenure	0
Balance	0
NumOfProducts	0
HasCrCard	0
IsActiveMember	0
EstimatedSalary	0
Exited	0

dtype: int64

```
print("Total Customers:", len(df))
```

Total Customers: 10000

```
active_churn = pd.crosstab(
    df['IsActiveMember'],
    df['Exited'],
    normalize='index'
)*100
```

```
print(active_churn)
```

Exited	0	1
IsActiveMember		
0	73.149103	26.850897
1	85.730926	14.269074

```
active_churn
```

```
df.groupby('Exited')['RSI'].mean()
```

		RSI	
Exited	0	73.149103	26.850897
	1	85.730926	14.269074

dtype: float64

```
import os
print(os.listdir())
```

```
['.config', 'log.txt', 'app.py', 'sample_data']
```

```
import os
print(os.listdir())
```

```
['.config', 'log.txt', 'app.py', 'European_Bank.csv', 'sample_data']
```

```
import pandas as pd

df = pd.read_csv("European_Bank.csv")

print(df.shape)
```

```
(10000, 14)
```

```
print(df.shape)
```

```
(10000, 14)
```

✓ Key Findings